

Analyzing Heart Signals Using Bayesian Inference

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Purpose

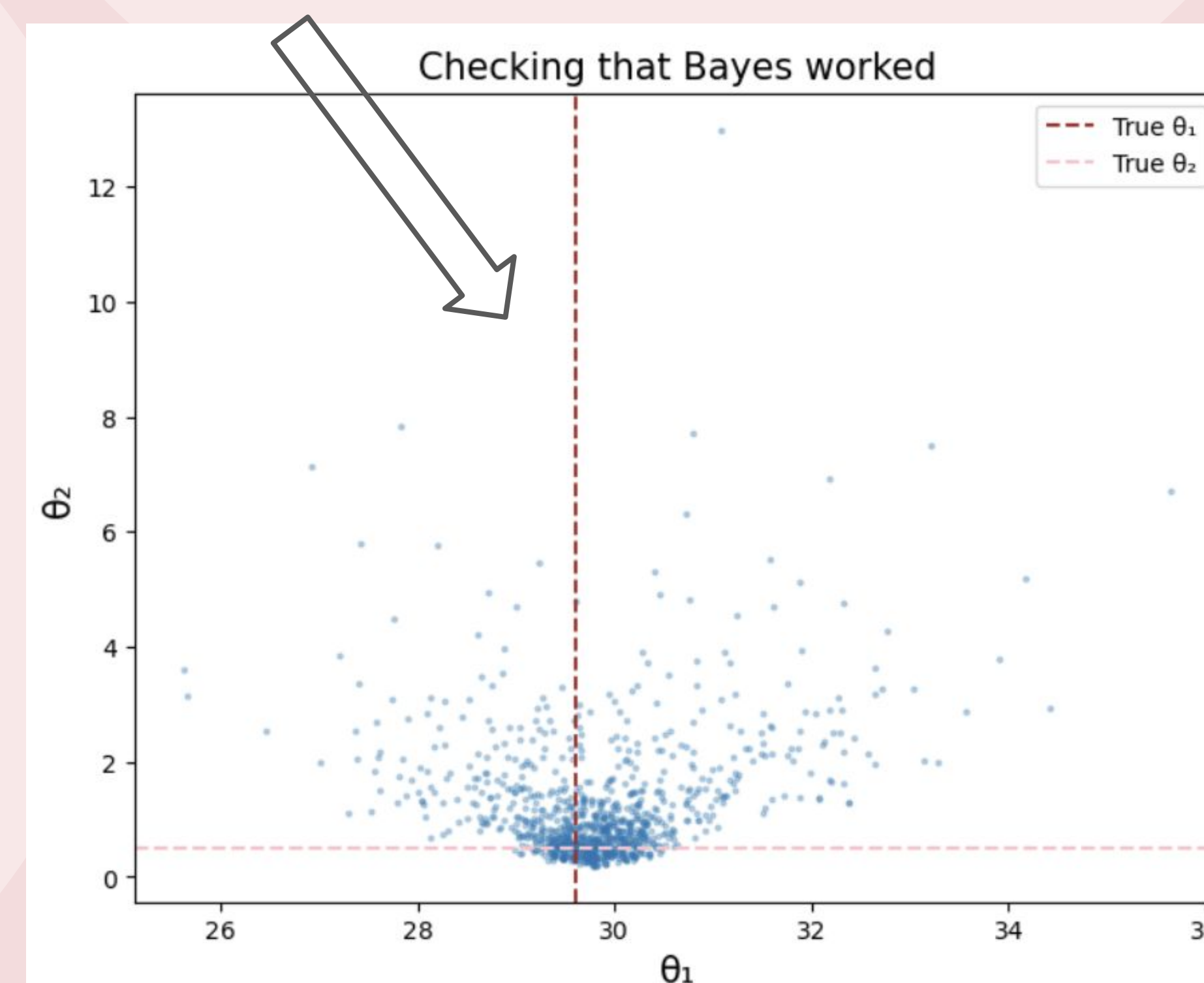
- Create a universal algorithm for analyzing heart signals – including for patients with heart disorders, such as arrhythmia
- Identify relationship between heartbeat timing and regularity

Goal

- Create an algorithm that successfully identifies peaks in ECG and PPG signals to analyze heart signals
- Establish a confident prior for our heartbeat timing utilizing Bayesian Inference
- Test our algorithms effectiveness in identifying correct peaks as compared to existing algorithms
- Develop a system of measuring quality of peak detection algorithms

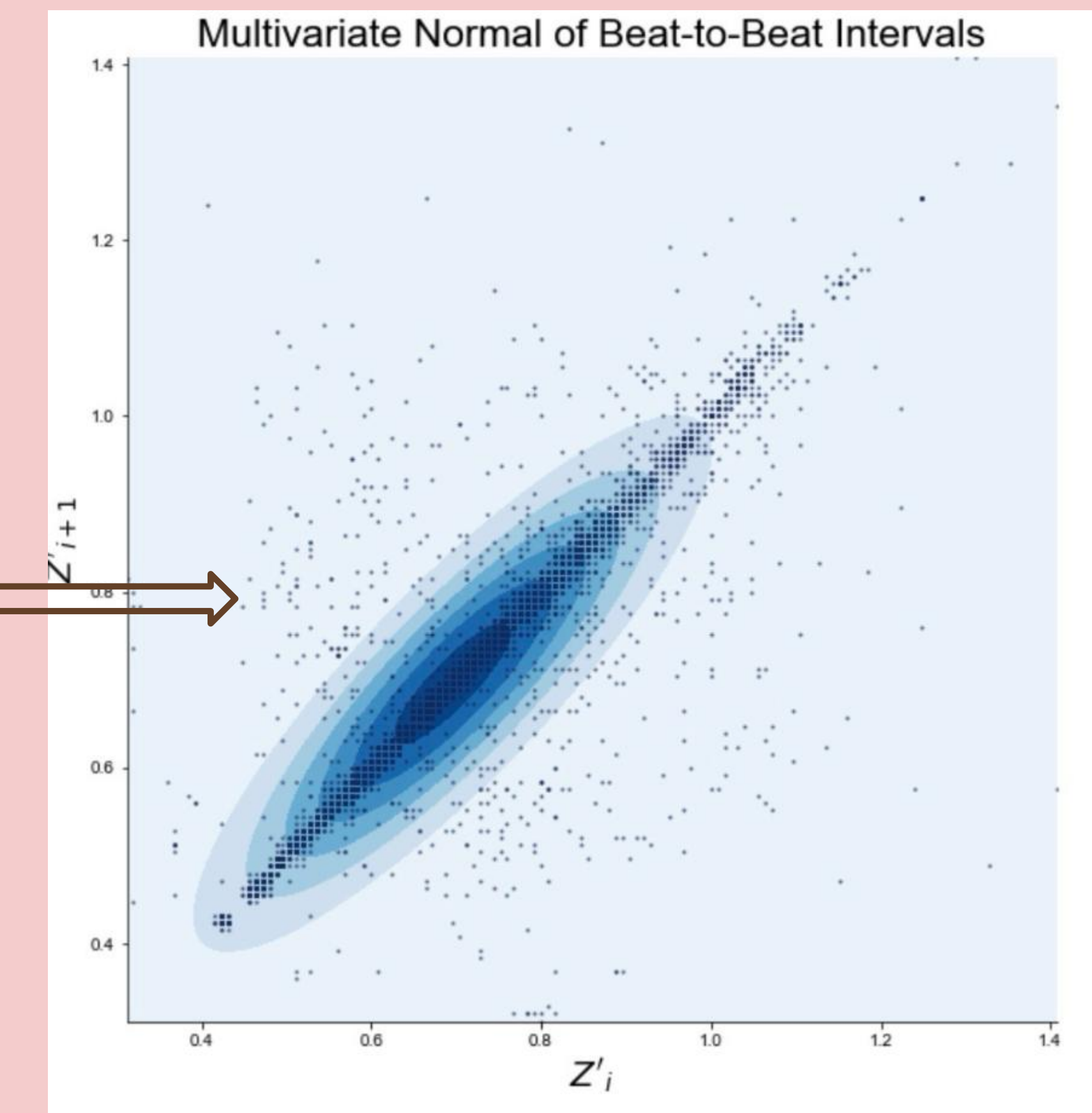
Bayesian Inference

The posterior $P(A|B)$ distribution represents our updated beliefs about our unknown parameter after having observed our given data (true values). To show that my algorithm for calculating Bayes successfully works, I created random “true” values and verified that the highest density of points sampled from the posterior were close to the “true” values.



Methods

- Testing various python packages, attempting to create my own algorithm for identifying peaks
- Using plots to predict likelihood between two heartbeats
- Testing our algorithm on critical patient datasets – for example, patients with arrhythmia
- Proof of concept of Bayes Rule with random values



Bayesian Inference Formula

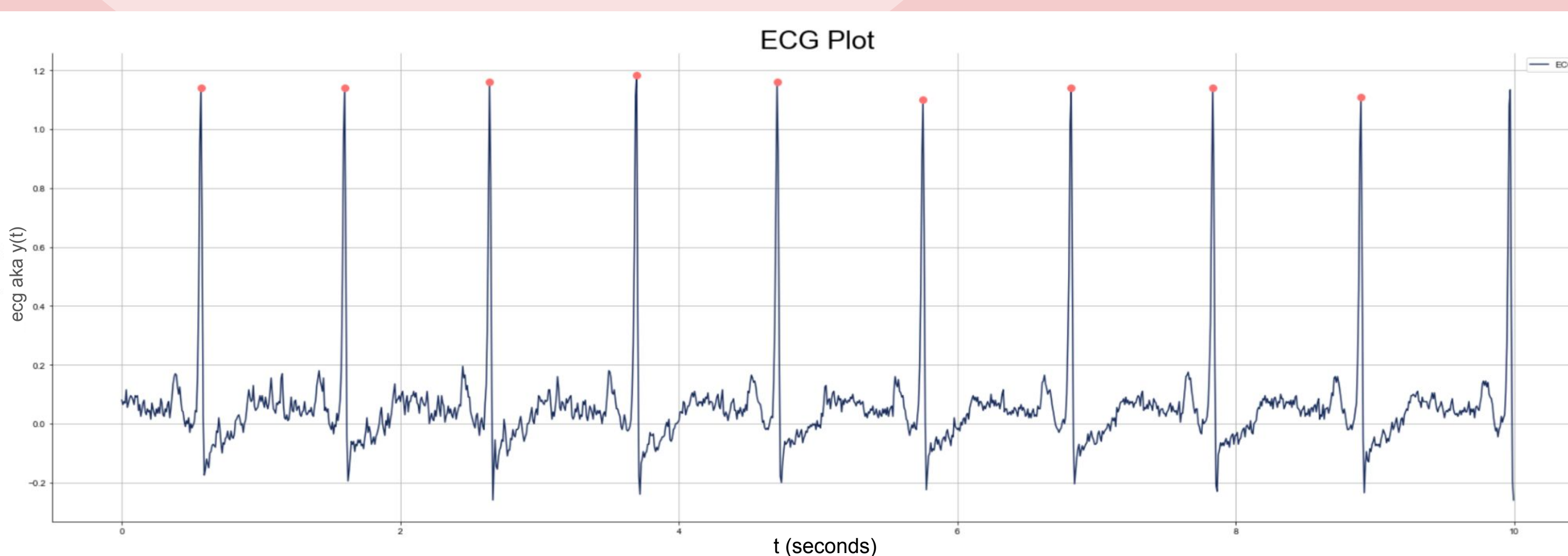
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Algorithm Quality Measurement

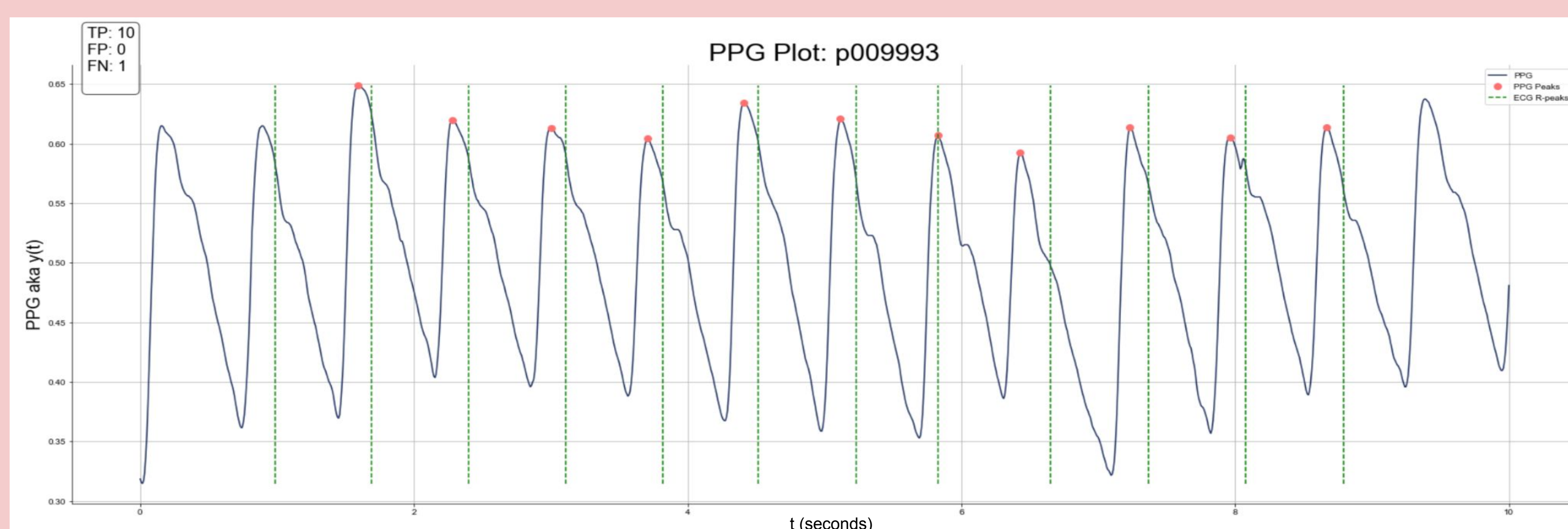
- Created four categories for measuring quality of algorithms
- FP = False Positive, FN = False Negative, TP = True Positive
- Continuous difference between heartbeats
- Began implementing this measurement on different peak detection methods (3)

Results

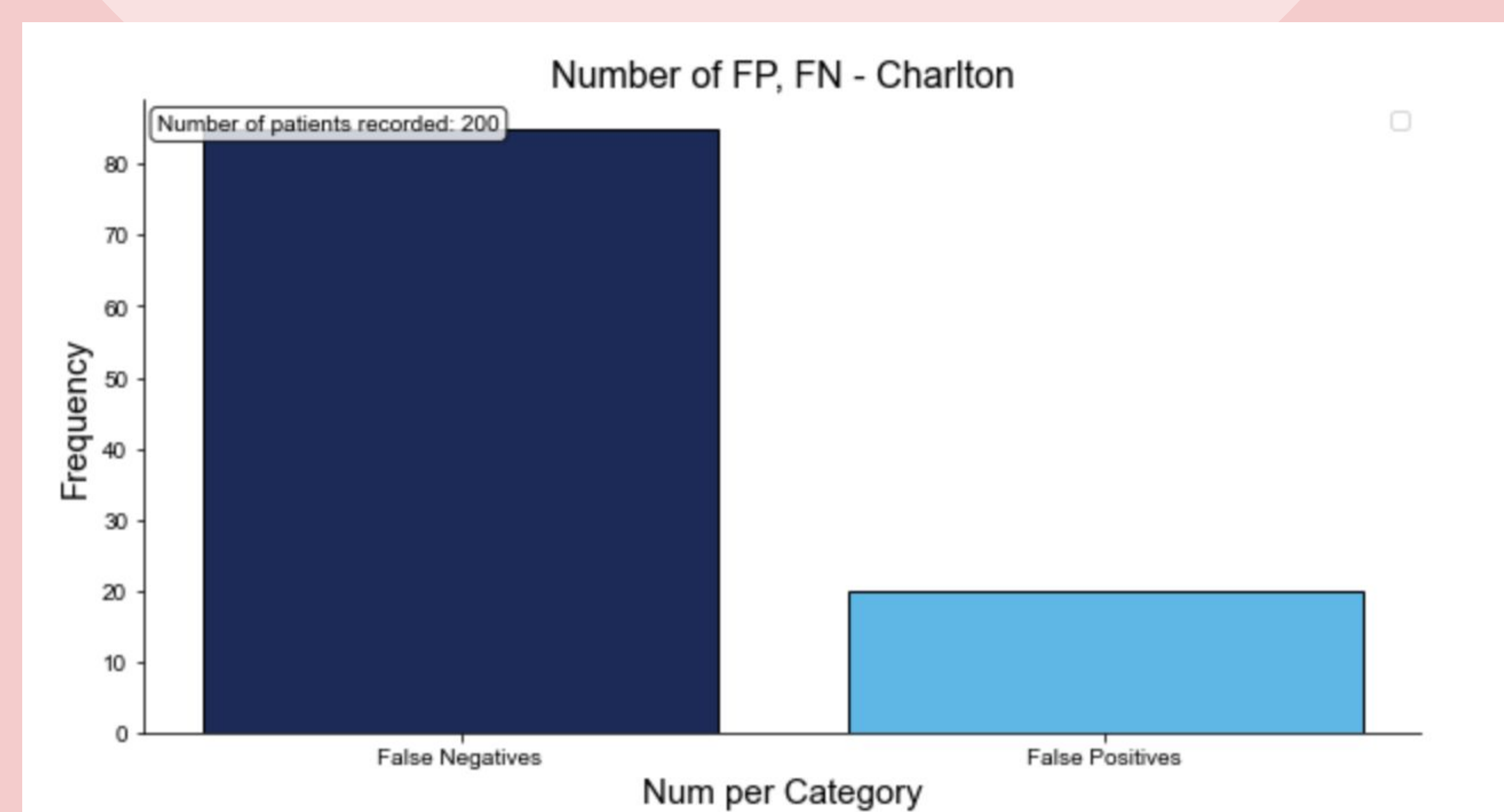
- Succeeded in creating an algorithm that detects and plots peaks of ECG and PPG signals (1A, 1B)
- Identified strong relationship between two successive peaks/heartbeats (2)
- Successfully created an algorithm that produces a posterior distribution that agrees with our true value
- Successfully created a system of measuring the quality of peak detection algorithms



1A



1B



3

